The Extension Problem for Orthomodular Observation Algebras

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1 Background

All measurement is finite. An experimenter performs finitely many trials, records finitely many digits, distinguishes finitely many outcomes. Yet the mathematical frameworks we use to organise empirical knowledge — probability theory, quantum mechanics, statistical mechanics — rest on infinite structures: σ-algebras, Hilbert spaces, continuous symmetry groups. The passage from finite observation to infinite structure is not automatic; it requires a commitment that no finite body of evidence can compel.

In probability theory, the locus of this commitment is *countable additivity.* Given a Boolean algebra $\mathcal{E}$ of events and a finitely additive charge $\ell : \mathcal{E} \to [0,1]$ with $\ell(\Omega) = 1$, the condition

$$E_n \downarrow \varnothing \quad \Longrightarrow \quad \ell(E_n) \to 0 \tag{1}$$

— that mass cannot persist along a vanishing sequence of events — is not derivable from any finite collection of observations. De Finetti [7] argued that only finite additivity is empirically grounded; Kolmogorov [14] adopted σ-additivity as an axiom and built modern probability on it. The question of *when* and *why* the passage from one to the other is justified has never been fully resolved.

In quantum theory, the same tension appears in a different guise. The algebra of propositions about a quantum system is not Boolean: incompatible observables cannot be simultaneously determined. The closed subspaces of a Hilbert space H form an orthomodular lattice $L(H)$ — a structure in which complementation is well-defined but distribution of meet over join may fail:

$$a \wedge (b \vee c) \;\neq\; (a \wedge b) \vee (a \wedge c) \qquad \text{in general.} \tag{2}$$

A *state* on such a lattice is a function $s : L(H) \to [0,1]$ satisfying

$$s(\mathbf{1}) = 1, \qquad s(a \vee b) = s(a) + s(b) \quad \text{whenever } a \perp b. \tag{3}$$

The non-distributivity means that s is additive only on orthogonal pairs, not on arbitrary finite unions. The classical machinery for extending finitely additive assignments to σ-additive ones — Carathéodory's outer-measure construction, which requires a Boolean algebra of sets — does not apply.

For the specific lattice $L(H)$, this difficulty is resolved by a remarkable rigidity. Gleason's theorem [12] shows that, when $\dim H \geq 3$, every state has the form

$$s(P) \;=\; \mathrm{tr}(\rho\, P) \tag{4}$$

for a unique density operator ρ, and is therefore automatically σ-additive. The geometry of Hilbert space is so constraining that the problem of the infinite simply does not arise — states have no room to misbehave.

But Hilbert space is very special. Orthomodular lattices arise in many other settings — the projection lattices of operator algebras, the logics of quantum systems with superselection rules, and the abstract lattices studied in quantum logic — and for these, no analogue of Gleason's rigidity is known. The question that probability theory and quantum theory share — when does finite coherence force countable behaviour? — remains open in this broader setting, and is the subject of the present note.

The programme "Structure from Observation" [16] approaches this question from a particular direction: rather than beginning with a pre-given space Ω of outcomes and asking which σ-algebras it supports, it begins with the *observations themselves* — a directed system of event algebras $\mathcal{E}_i$ with compatible charges ℓ_i — and asks what the directed structure alone determines.

Paper I answers this for Boolean event algebras. A compatible family of finitely additive charges extends to a σ-additive probability measure if and only if collective exhaustion (CE) holds — condition (1) applied uniformly across the directed system. Two constructions realise the extension:

- **Carathéodory:** assumes CE; produces a measure on $(\Omega, \sigma(\mathcal{C}))$ directly.
- **Stone:** assumes nothing; produces a Baire measure on the Stone space $\mathrm{St}(\mathcal{C})$ unconditionally.

Under discriminability, the two agree. Three questions orient the broader programme:

(1) What must be added to observation to get structure?

(2) Reconstruction without points: what structure is already present in the relational description — contexts, directed refinement — before point-spaces are assumed?

(3) Local-to-global extension: what conditions on the *site* control whether local data extend globally?

Paper I resolves (2) and (3) for Boolean algebras. The present note asks whether any of this survives when the event algebra is orthomodular but not distributive.

Epperson and Zafiris [10] advocate a *relational realist* ontology in which structure is constituted by relations among contexts, not by points in a pre-given space. An orthomodular lattice A is the natural algebraic expression of this: its elements are propositions, its partial order is entailment, and its non-distributivity reflects the impossibility of assigning simultaneous truth values to all propositions. The dual space $S_0(A)$ of McDonald–Bimbó [17] is built from *filters* — consistent collections of propositions — not from points of a pre-existing space. This is question (2) applied to non-Boolean algebras: what structure does the relational description already contain?

In the Boolean case, one might hope that the passage from finite to countable additivity could be understood as a *sheaf condition* — local states gluing to a global σ-additive measure. Biesel [1] shows this hope is disappointed:

- Finitely additive charges form a sheaf for finite covers.
- σ-additive measures form a sheaf for countable covers.
- The step between is the *definition* of σ-additivity, not a structural theorem.

This is the boundary at which the present problem begins.

Paper II [15] introduces three positions on the relationship between observation and reality, each a constraint on where the dual-space measure lives:

Empirical adequacy (EA)

The dual-space measure agrees with all observations.

Probabilistic realism (PR)

The measure descends to a realisation space.

Value-definite realism (VDR)

A global two-valued homomorphism exists.

These are nested: VDR $\Rightarrow$ PR $\Rightarrow$ EA. For Boolean algebras, all three are commensurable — every EA-theory admits a VDR-completion. For OMLs with dim ≥ 3, Gleason's theorem (4) rescues PR, but Kochen–Specker [13] blocks VDR. The extension problem sits at the EA $\to$ PR transition: can we pass from a state on the algebra (EA) to a σ-additive measure on the dual space (PR)? If so, under what conditions — and is descent then a free choice (as in the Boolean case) or a structural consequence of the algebra and state together?

Van Fraassen's distinction [22] between empirical adequacy and structural commitment, which Paper I makes precise via the Stone construction, thus acquires a sharper form in the OML setting: the Stone route is no longer available, the Carathéodory route is blocked by non-distributivity, and the passage from coherent local data to global probability — question (3) — requires genuinely new ideas.

2 The problem

Let A be an orthomodular lattice (OML) and let $s : A \to [0, 1]$ be a state (3). The McDonald–Bimbó duality [17] associates to A a compact topological space

$$S_0(A) \;=\; \big(F(A),\, \subseteq,\, \perp_A,\, P(A),\, T(S)\big),$$

where $F(A)$ is the set of filters on A and $P(A) \subseteq F(A)$ is the set of prime filters. For each $a \in A$, set

$$h(a) \;=\; \{x \in F(A) : a \in x\}.$$

The map h is an OML isomorphism from A onto the $\perp$-stable clopen sets $\mathrm{CO}(S_0(A))^\dagger$. Define a set function μ on these clopens by

$$\mu\big(h(a)\big) \;=\; s(a). \tag{5}$$

Question 2.1 (Extension)**.** Under what conditions on the OML A does μ extend to a σ-additive measure on the Baire (or Borel) σ-algebra of $S_0(A)$?

This is the direct OML analogue of the Stone extension theorem established for Boolean algebras in [16].

In the Boolean case, the extension is automatic. The clopens $\mathrm{Clop}(\mathrm{St}(\mathcal{C}))$ form a Boolean algebra, the state s restricts to a finitely additive charge on this algebra, and the Carathéodory extension theorem (or Choksi's projective-limit argument [6]) produces the σ-additive extension. Compactness of the Stone space gives σ-subadditivity for free.

In the OML case each of these steps breaks:

(i) $\mathrm{CO}(S_0(A))^\dagger$ is an OML, *not* a Boolean algebra.

(ii) The state s is additive on orthogonal pairs only, not on arbitrary finite unions.

(iii) **Carathéodory does not apply** — there is no Boolean algebra of sets on which to run the outer-measure construction.

(iv) Compactness holds (McDonald–Bimbó, Lemma 3.5 [17]), but the σ-subadditivity argument requires orthogonality structure that the topology alone does not provide.

The gap is sharp: "finitely additive on a compact Boolean algebra of clopens" automatically extends; "orthogonally additive on a compact OML of clopen $\perp$-stable sets" does *not*.

For the prototypical case $A = L(H)$ with $\dim H \geq 3$, Gleason's theorem (4) resolves the problem completely: every state is induced by a density operator, and the resulting measure is σ-additive on $L(H)$. The extension to $S_0(L(H))$ follows. For general OMLs, no Gleason-type representation is available. Question 2.1 asks for the structural conditions that substitute for the Hilbert-space geometry.

Feature	**Boolean (Paper I)**	**OML (this problem)**
Dual space	$\mathrm{St}(A)$ (ultrafilters)	$S_0(A)$ (all filters)
Distinguished subset	pure(Ω), not canonical	$P(A)$, canonical
Clopens	Boolean algebra	OML (non-distributive)
Additivity of s	full (finite)	orthogonal pairs only
Extension	automatic (Carathéodory)	requires Gleason-type result
Realisation constrained?	No	Yes (Kochen–Specker [13])

Table 1: Structural comparison of the extension problem.

Two axes: extension and descent. The passage from a state on A to a measure on $S_0(A)$ splits into two axes that the Boolean case fuses but the OML case separates. Since $S_0(A)$ is a Stone space, its *full* clopen algebra $\mathrm{CO}(S_0(A))$ is Boolean, whereas the state lives only on the $\perp$-stable clopens $\mathrm{CO}(S_0(A))^\dagger$ — the OML, the image of h. These share only meet; join, complement, and bottom differ.

Extension Extend μ from the $\perp$-stable clopens to a finitely additive charge on the *full* Boolean algebra of clopens. This requires consistent values on non-$\perp$-stable clopens — unions $h(a) \cup h(b)$ for non-orthogonal a, b, which lie outside $\mathrm{im}(h)$ whenever A is non-Boolean. This is the Pták–Pulmannová frontier; σ-additivity of the state does not help here.

Compactness Once extension succeeds, the charge extends to a σ-additive Borel measure automatically. Unconditional.

In the Boolean case (Paper I), the extension axis is *free*: finite additivity on the clopen Boolean algebra plus compactness gives the Stone measure unconditionally — "the first arrow is free." In the OML case, **the extension axis is no longer free** — it is the open problem. The non-distributivity that blocks Carathéodory is exactly the obstruction to extending past the orthogonal pairs.

Descent. σ-additivity of the state governs descent, not extension. Even once μ extends to a measure $\hat{\mu}$ on $S_0(A)$, the *descent* question — whether $\hat{\mu}$ concentrates on the "physical" points — changes character:

Boolean Does $\hat{\mu}$ concentrate on pure(Ω)? If and only if the charges are σ-additive. The choice of realisation space Ω is unconstrained.

OML Does $\hat{\mu}$ concentrate on $P(A)$? Here $P(A)$ is part of the structure, not a free choice. The Kochen–Specker obstruction [13] prevents simultaneous realisation of all filters; the Bub–Clifton theorem [2] shows that the state determines a maximal Boolean subalgebra of definite values.

In the OML setting, the algebra and state *together* constrain which filters are "realised." Descent is not a geometric commitment but a structural consequence.

Remark 2.2 (Decomposition lives on the descent axis)**.** De Simone–Navara [8] decompose OMP states as $s = s_\sigma + s_{\mathrm{wpfa}}$ (a σ-additive part and a weakly purely finitely additive part), the orthomodular analogue of Yosida–Hewitt. By the axis split above, the wpfa component governs the *descent* axis — it is the analogue of the purely finitely additive part ℓ_p in the Boolean case — and *not* the extension axis. A natural-looking conjecture, "s extends iff $s_{\mathrm{wpfa}} = 0$," is therefore mis-targeted: vanishing of s_{wpfa} concerns whether the (already σ-additive) state's measure concentrates on $P(A)$, while extension is blocked by non-distributivity regardless of the wpfa component. The decomposition theory and the extension problem live on different axes.

A caveat compounds this. The McDonald–Bimbó duality [17] is *finitary*: the identity $h(\bigvee_n a_n) = (\bigcup_n h(a_n))^{\perp\perp}$ holds for finite joins but fails in general for countable joins, since finitary OML homomorphisms need not preserve infinite suprema. Making even the descent argument rigorous — the analogue of Paper I's "σ-additivity $\Leftrightarrow$ concentration" — requires σ-completeness and continuity hypotheses the 2023 duality does not supply, and is itself open.

3 What is known

In the Boolean case, the Kelley–Vladimirov–Pták characterisation (Fremlin, Theorem 391D [11]) gives a complete algebraic answer:

Theorem 3.1 (KVP)**.** *A Boolean algebra B carries a strictly positive σ-additive measure if and only if B is*

(a) Dedekind σ-complete,
(b) weakly (σ, ∞)-distributive, and
(c) chargeable (carries a strictly positive finitely additive measure).

Each condition is algebraic and non-trivially constraining. For OMLs, the analogues fail or degenerate:

(a) Dedekind σ-completeness makes sense but is very strong.
(b) Weak (σ, ∞)-distributivity has no clean OML analogue — the definition relies on distributivity of $\bigwedge$ over $\bigvee$, which may fail orthomodularly.
(c) Chargeability has no known structural characterisation on OMLs; the existence of a strictly positive finitely additive measure cannot be read off from the lattice structure.

The positive results that do exist for OMLs all require algebraic structure rich enough to approximate Hilbert-space geometry:

Gleason [12] $A = L(H)$, $\dim H \geq 3$. Every state has the form (4); σ-additivity is automatic.

Bunce–Wright [3] A the projection lattice of a JBW-algebra. σ-additivity via operator-algebraic structure.

Chetcuti–Dvurečenskij [5] Lattice effect algebras with completeness conditions close to von Neumann algebras.

The obstructive results show that the Boolean extension machinery cannot be transplanted:

Pták–Pulmannová [20]
If every unital subadditive measure on an orthomodular poset is a state, the poset must be Boolean. This is the structural reason no KVP analogue exists for OMLs: conditions strong enough to force σ-additivity collapse the lattice back to a Boolean algebra.

Navara [18] Regularity does *not* force σ-additivity on general orthomodular posets. The Béaver–Cook result is special to von Neumann algebras.

Pták [19] Exotic OML state spaces exist; the obstruction is combinatorial (Greechie pasting), not from the centre.

Recent incremental work confirms the field is active but no one has found the right condition: Voráček–Pták [23] study signed measures on orthomodular posets; Burešová–Pták [4] investigate variations on regularity conditions; De Simone–Navara study Yosida–Hewitt-type decompositions for orthomodular posets.

The gap between "too weak" (allows non-σ-additive states) and "too strong" (collapses to Boolean) appears structurally robust. This is not a problem where further reading will produce the answer — it requires a new idea.

All known exotic OML state spaces are built by **Greechie pasting** [24, 21, 19]. Progress on the extension problem likely requires either:

(a) a new pasting technique that controls σ-additivity, or

(b) an approach that bypasses the state space entirely (e.g., categorical or topos-theoretic [9]).

4 Assessment

Question 2.1 is a precise, well-formulated problem. A positive answer would unify three results under a single framework — states on OMLs → measures on dual spaces → constrained descent:

(i) Paper I [16]: the Boolean case.
(ii) Gleason's theorem [12]: the $L(H)$ case.
(iii) The Bub–Clifton theorem [2]: descent and realisation.

Difficulty. High. The Pták–Pulmannová obstruction shows this is not merely "find the right condition" — conditions strong enough to force extension destroy the non-Boolean structure.

Novelty. The formulation via McDonald–Bimbó duality [17] appears to be new. The Döring–Isham topos approach [9] addresses related questions but via presheaves, not the filter-space duality.

Status. Open but structurally resistant. Not a current active lead — needs collaboration or a new idea.

References

[1] Owen Biesel. Sheaves of probability. arXiv preprint arXiv:2401.01968, 2024.

[2] Jeffrey Bub and Rob Clifton. A uniqueness theorem for 'no collapse' interpretations of quantum mechanics. *Studies in History and Philosophy of Modern Physics*, 27(2):181–219, 1996.

[3] Leslie J. Bunce and J. D. Maitland Wright. The Mackey–Gleason problem. *Bulletin of the American Mathematical Society*, 26(2):288–293, 1992.

[4] Klára Burešová and Pavel Pták. Variations on regularity conditions for orthomodular posets. 2023. Preprint.

[5] Emmanuel Chetcuti and Anatolij Dvurečenskij. On the Gleason theorem for effect algebras. *Mathematica Slovaca*, 55(3):297–308, 2005.

[6] J. R. Choksi. Inverse limits of measure spaces. *Proceedings of the London Mathematical Society*, 8(3):321–342, 1958.

[7] Bruno de Finetti. *Theory of Probability*, volume 1. Wiley, 1974.

[8] Anna De Simone and Mirko Navara. Yosida–Hewitt and Lebesgue decompositions of states on orthomodular posets. *Journal of Mathematical Analysis and Applications*, 255(1):74–104, 2001.

[9] Andreas Döring and Chris J. Isham. A topos foundation for theories of physics: I. Formal languages for physics. *Journal of Mathematical Physics*, 49(5):053515, 2008.

[10] Michael Epperson and Elias Zafiris. *Foundations of Relational Realism: A Topological Approach to Quantum Mechanics and the Philosophy of Nature*. Lexington Books, Lanham, MD, 2013.

[11] D. H. Fremlin. *Measure Theory, Vol. 3: Measure Algebras*. Torres Fremlin, Colchester, 2004.

[12] Andrew M. Gleason. Measures on the closed subspaces of a Hilbert space. *Journal of Mathematics and Mechanics*, 6(6):885–893, 1957.

[13] Simon Kochen and Ernst P. Specker. The problem of hidden variables in quantum mechanics. *Journal of Mathematics and Mechanics*, 17(1):59–87, 1967.

[14] A. N. Kolmogorov. *Foundations of the Theory of Probability*. Chelsea, 1933.

[15] Patrick S. Mahon. Distributivity and the commensurability of empirical adequacy and realism. Preprint, 2026.

[16] Patrick S. Mahon. When does observational coherence determine probability? Preprint, 2026.

[17] J. McDonald and Katalin Bimbó. Duality for orthomodular lattices. 2023. Preprint.

[18] Mirko Navara. When is the integral on quantum probability spaces additive? *Bulletin of the Polish Academy of Sciences — Mathematics*, 40(1):29–33, 1992.

[19] Pavel Pták. Exotic logics. *Colloquium Mathematicum*, 54(1):1–7, 1987.

[20] Pavel Pták and Sylvia Pulmannová. *Orthomodular Structures as Quantum Logics*, volume 44 of *Fundamental Theories of Physics*. Kluwer, Dordrecht, 1991.

[21] Frederic W. Shultz. A characterization of state spaces of orthomodular lattices. *Journal of Combinatorial Theory, Series A*, 17(3):317–328, 1974.

[22] Bas C. van Fraassen. *The Scientific Image*. Oxford University Press, 1980.

[23] Václav Voráček and Pavel Pták. Signed measures on orthomodular posets. 2023. Preprint.

[24] Alexander Wilce. Test spaces. In Kurt Engesser, Dov M. Gabbay, and Daniel Lehmann, editors, *Handbook of Quantum Logic and Quantum Structures: Quantum Logic*, pages 443–549. Elsevier, 2009.